

SWETHA KAMBHAMPATI

Bengaluru, India

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Professional Summary

AI/ML Engineer with 3+ years of industry experience at Wipro and currently a Generative AI Intern at Wabtec while pursuing an M.Tech in Artificial Intelligence and Machine Learning at VIT Vellore. Hands-on experience in Large Language Models (LLMs), Amazon Bedrock, Model Context Protocol (MCP), Deep Learning, and AI-powered automation. Passionate about designing scalable GenAI applications and intelligent data-driven solutions.

Education

M.Tech – Artificial Intelligence & Machine Learning <i>VIT Vellore</i>	2025–Present <i>CGPA: 9.62</i>
B.Tech – Electrical and Electronics Engineering <i>Acharya Institute of Technology, Bengaluru</i>	2017–2021 <i>CGPA: 7.9</i>

Technical Skills

Programming Languages: Python, Java, SQL

ML & AI: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering, Fine-Tuning, Retrieval-Augmented Generation (RAG), Model Evaluation, Feature Engineering, Model Quantization

LLM & GenAI: Amazon Bedrock, Model Context Protocol (MCP), AI Agents, Prompt Compression, Prompt Optimization, Hugging Face Transformers

Deep Learning & Computer Vision: CNN, RNN, LSTM, YOLO (Object Detection), OCR

Libraries & Frameworks: TensorFlow, PyTorch, Keras, Hugging Face, Pandas, NumPy, Scikit-learn

Cloud & Data Engineering: AWS, Amazon Bedrock, Apache Spark (Academic), Databricks, Pentaho, SQL Database

Data Analytics & Visualization: Power BI, Excel, Predictive Analytics, Streamlit

Automation & Platforms: Power Automate, Power Apps, UiPath Orchestrator, AI Builder

Deployment & Integration: REST APIs, Docker, Kubernetes, Edge Intelligence

Tools: Jupyter Notebook, pgAdmin, SQLite, ServiceNow, ABBYY FlexiCapture

Work Experience

Generative AI Intern Wabtec Corporation, Bengaluru	June 2026 – Present
- Developing enterprise Generative AI applications using Amazon Bedrock, Large Language Models (LLMs), and the Model Context Protocol (MCP). - Building AI workflows by integrating LLMs with external tools and data sources to enable intelligent automation. - Collaborating with cross-functional teams to design, evaluate, and deploy scalable AI solutions.	
Project Engineer Wipro, Bengaluru	Sept 2021 – Oct 2024
- Developed enterprise automation solutions using Power Automate, Power Apps, and RPA, reducing manual effort by approximately 30%. - Managed the end-to-end automation lifecycle, including requirement gathering, development, testing, deployment, and SQL-based data validation.	

Projects

- Software Requirement Classification using LLM Fine-Tuning**
Developed an NLP pipeline using Gemini, transformer embeddings, and LoRA fine-tuned Mistral-7B for software requirement classification, improving performance by 20% over baseline models.
- Label Uncertainty Handling in Autonomous Driving using GenAI and XAI (Ongoing)**
Developing a CNN-based framework leveraging Generative AI and Explainable AI (XAI) to improve robustness against noisy and ambiguous labels in autonomous driving datasets.
- Cross-View Maize Disease Detection and Severity Estimation (Ongoing)**
Building a PyTorch-based computer vision pipeline for crop disease detection and lesion-based severity estimation with cross-view generalization from leaf to canopy imagery.
- Stock Market Prediction and Forecasting**
Implemented ARIMA and machine learning models for financial time-series forecasting with data preprocessing, feature engineering, and model evaluation.

Certifications

- PG Level Certification in Business Analytics**, Indian Institute of Science (IISc) **Aug 2024 – Jul 2025**